

LSH Banding

* Indicates required question

1. Email *

2. Roll number *

3. Password *

Question

Suppose we have computed signatures for a number of columns, and each signature consists of 24 integers, arranged as a column of 24 rows. There are N pairs of signatures that are 50% similar (i.e., they agree in half of the rows). There are M pairs that are 20% similar, and all other pairs (an unknown number) are 0% similar.

We can try to find 50%-similar pairs by using Locality-Sensitive Hashing (LSH), and we can do so by choosing bands of 1, 2, 3, 4, 6, 8, 12, or 24 rows. Calculate approximately, in terms of N and M , the number of false positive and the number of false negatives, for each choice for the number of rows. Then, suppose that we assign equal cost to false positives and false negatives (an atypical assumption). Which number of rows would you choose if $M:N$ were in each of the following ratios: 1:1, 10:1, 100:1, and 1000:1? Identify the correct choices from the list below.

4. Options

1 point

Check all that apply.

- ☐ For $M = 1000N$, pick $r = 8$.
- ☐ For $M = N$, pick $r = 3$.
- ☐ For $M = N$, pick $r = 2$.
- ☐ For $M = 1000N$, pick $r = 4$.